

EXPERIMENT 23: PERFORM SHARPENING OF IMAGE USING UNSHARP MASKING

Aim

To sharpen an image using the Unsharp Masking technique in Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image to grayscale.
4. Blur the image using Gaussian Blur.
5. Combine the original and blurred images using `addWeighted()`.
6. Display and save the sharpened image.

Formula

Sharpened Image = Original Image - Blurred Image

Python Code

```
import cv2

img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

blur = cv2.GaussianBlur(gray, (9, 9), 2)

sharpened = cv2.addWeighted(gray, 1.5, blur, -0.5, 0)

cv2.imwrite("unsharp_mask_output.jpg", sharpened)

cv2.imshow("Original Image", gray)
```

```
cv2.imshow("Sharpened Image", sharpened)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the image was successfully sharpened using the Unsharp Masking technique.